

Clinical Risk Assessment Report: Patient 58

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.53x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 1.6%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.19 [Avg (59th %ile)] (+1.5%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 72.0 [Top 18%] (-1.6%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.21 [Avg (47th %ile)] (-1.6%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.48 [Avg (34th %ile)] (-1.3%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -0.49 [Bottom 25%] (-1.0%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 1.1 [Top 13%] (-0.9%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.12 [Avg (57th %ile)] (-0.7%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: -0.1 [Avg (40th %ile)] (-0.4%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)

